

BEAR: A Unified Framework for Evaluating Relational Knowledge in Causal and Masked Language Models

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Abstract

Knowledge probing assesses to which degree a language model (LM) has successfully learned relational knowledge during pre-training. Probing is an inexpensive way to compare LMs of different sizes and training configurations. However, previous approaches rely on the objective function used in pre-training LMs and are thus applicable only to masked or causal LMs. As a result, comparing different types of LMs becomes impossible. To address this, we propose an approach that uses an LM’s inherent ability to estimate the log-likelihood of any given textual statement. We carefully design an evaluation dataset of 7,731 instances (40,916 in a larger variant) from which we produce alternative statements for each relational fact, one of which is correct. We then evaluate whether an LM correctly assigns the highest log-likelihood to the correct statement. Our experimental evaluation of 22 common LMs shows that our proposed framework, BEAR, can effectively probe for knowledge across different LM types. We release the BEAR datasets and an open-source framework that implements the probing approach to the research community to facilitate the evaluation and development of LMs.

1 Introduction

Pre-trained language models (LMs) are the backbone of current state-of-the-art NLP approaches. A key property is the syntactic and semantic knowledge stored in their internal parameters, allowing them to generalize beyond given training data when fine-tuning for a specific downstream NLP task. Due to their importance and the large number of proposed LMs, prior work has sought to improve the ability to measure the amount of factual knowledge encoded in LMs, thereby facilitating the comparison of different LMs (Petroni et al., 2019; Po-

The capital of France is [MASK].

(a) LAMA probe: Single-subtoken mask prediction

The capital of Uganda is Thimphu.

... Kampala.

... Buenos Aires.

... Bandar Seri Begawan.

(b) BEAR probe: Rank answer options of arbitrary length

Figure 1: Comparison of the LAMA and BEAR probes. Both probes query LMs given a template (here in black), the subject of the relation (blue), and the object (orange). LAMA masks the object and predicts a single token as the answer. In BEAR, we create separate textual statements for a set of potential answers and select the statement with the highest (pseudo) log-likelihood as assigned by the LM. This method allows us to include multi-token answers and evaluate causal and masked LMs.

erner et al., 2020; Cao et al., 2021; Kalo and Fichtel, 2022).

The LAMA probe (Petroni et al., 2019) is the seminal work in studying commonsense and relational knowledge in LMs and is widely used for inexpensive evaluation and model comparison (see Youssef et al. (2023) and Cao et al. (2023) for an overview). The idea is to use relational knowledge from an existing knowledge base (KB) and create cloze-style statements for an LM to fill in.

For instance, the entities “France” and “Paris” may be connected through the HAS-CAPITAL relation in a given KB, indicating that Paris is the capital of France. From this, LAMA constructs the sentence “The capital of France is [MASK]” and evaluates whether an LM predicts the correct token to complete this factual sentence. LAMA, therefore, effectively reuses the masked language modeling objective of the bidirectional family of LMs (Devlin et al., 2019) to probe for knowledge. This example is shown in Figure 1a.

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Limitations of LAMA. While LAMA offers a straightforward approach to probing, it also has significant limitations.

First, LAMA requires the correct answer to be part of the evaluated LM’s subtoken vocabulary, restricting the relational knowledge that can be tested to single-subtoken answers (such as “Paris” in Figure 1a). Thus, it cannot test for relational facts with long or rare answers (as shown in Figure 1b).

Second, and most importantly, LAMA relies on the masked language modeling objective. This makes LAMA inapplicable for LMs trained with other objectives. Therefore, it excludes causal LMs such as the GPT-family of models (Radford et al., 2019). To the best of our knowledge, no factual knowledge probe currently exists that applies to both masked and causal LMs.

Third, various prior works have noted limitations of the relational data used in the LAMA probe, such as (1) a heavily skewed answer space, favoring some answers over all others (Jiang et al., 2020b; Zhong et al., 2021; Cao et al., 2021), (2) overly revealing entity names (Poerner et al., 2020), (3) and issues involving knowledge with multiple correct answers, causing correct answers to be counted as errors (Kalo and Fichtel, 2022).

These limitations of both the probing approach and the data impair LAMA’s ability to measure and compare the relational knowledge of different LMs accurately.

Contributions. To address these issues, we propose BEAR, a unified knowledge probe for both causal and masked LMs. Instead of casting the evaluation as a token prediction problem over the entire vocabulary of an LM, we present a set of answer options for each relation instance, create a textual statement for each option, and leverage the inherent ability of each LM to assign a log-likelihood score to statements, thereby ranking these options. See Figure 1b for an illustration.

We argue that this approach has numerous benefits in that it (1) allows us to evaluate both masked and causal LMs, (2) imposes no restrictions on the answer space, (3) allows us to design a new evaluation dataset that addresses a range of issues such as answer skews and multiple correct answers noted in prior work. In more detail, our contributions are:

1. We present an analysis of the weaknesses of the LAMA probe and follow-up works to derive desiderata for the BEAR probe (see Section 2).

2. We propose to query knowledge as a multiple-choice selection problem in which an LM estimates the (pseudo) log-likelihood of a given answer template with each choice filled in (see Section 3).
3. We construct a novel evaluation dataset that reflects the desiderata identified in our analysis (see Section 4).
4. We use BEAR to evaluate a range of common masked and causal LMs (see Section 5).

To enable the community to employ the proposed probing method and dataset, we publicly release¹ the evaluation framework *lm-pub-quiz* (based on the *minicons*, Misra, 2022) as well as the dataset *BEAR*².

2 LAMA and Follow-Up Work

We discuss the technical details of the LAMA probe first, followed by an analysis of its weaknesses.

LAMA evaluation data. The LAMA benchmark was originally composed of four separate datasets named after their respective sources: SQuAD (Rajpurkar et al., 2016), GoogleRE³, ConceptNET (Speer and Havasi, 2012), and T-REx (Elsahar et al., 2018). However, subsequent research mainly focused exclusively on the T-REx subset. Its knowledge base comprises 41 *relations* derived from Wikidata. Each relation contains at most 1,000 *relation instances* in the form of subject-relation-object triples $\langle s, r, o \rangle$ where s is the subject (e.g., “France”), r the relation (e.g., HAS-CAPITAL), and o an object (e.g., “Paris”).

There are three types of relations in LAMA: 1-1 (one-to-one, e.g., HAS-CAPITAL), N-1 (many-to-one, e.g., HAS-LANGUAGE), and N-M (many-to-many, e.g., SHARES-BORDER-WITH). Relations of the N-1 type allow multiple subjects to relate to one object, while the latter permits many subjects to be associated with numerous objects.

Relation identifiers. All relations are linked to a corresponding relation in Wikidata and thus have

¹The library, the probing dataset, as well experimental artifacts can be retrieved via the following URL: <https://lm-pub-quiz.github.io/>

²Benchmark for Evaluating Associative Reasoning), CC BY-SA license.

³<https://code.google.com/archive/p/relation-extraction-corpus/>

unique IDs. For instance, the CAPITAL-OF relation in LAMA corresponds to Wikidata relation P1376 (see Table 1 for more examples). It facilitates comparison across different datasets since all follow-up works to LAMA, including BEAR, derive their relations from Wikidata.

Templates. Each relation in LAMA has a textual template with placeholders for subject and object. For CAPITAL-OF, the template is “[X] is the capital of [Y].”, where [X] is a placeholder for the subject, while [Y] is the placeholder for the object. At test time, the subject of a given relation is filled in the template, while the object is replaced by a [MASK]-token. This procedure results in a masked sentence (e.g., “Paris is the capital of [MASK].”) for which the LM is tasked to predict the masked token.

2.1 Issue 1: Single Subtoken Answers

As noted by Petroni et al. (2019), LAMA is restricted to single-subtoken answers for factual knowledge queries. This limitation causes issues as LMs split most words into multiple subtokens, and most LMs differ in how they perform the splits. To illustrate, consider how the country name “Togo” is tokenized by different versions of BERT: the bert-base-cased model splits the word into two subtokens ([To, ##go]), whereas the bert-base-uncased variant preserves it as a single subtoken ([togo]).

An analysis of 194 UN member country names is an excellent example of how such a restrictive condition affects the size of a hypothetical dataset. When restricting answer space to single tokens, 32% and 27% of available country names would have to be discarded for cased and uncased versions of BERT, respectively. Worse, the RoBERTa models (Liu et al., 2019) that uses a BPE-based tokenizer would split 88% of all country names. Refer to Table 1 for a list of how many LAMA instances need to be discarded when evaluating xlm-roberta-base (Conneau et al., 2020) and roberta-base models.

Comparison of different LMs. Because the tokenizer bundled with each model differs, comparing various LMs becomes only possible if the models tokenize the answers in the same way. To address this, practitioners are currently reverting to using the intersection of single-token vocabularies derived from all LMs being compared. However, in practice, this further limits the scope of relational knowledge that can be included in the evaluation.

ID	Relation	xlm-roberta-base	roberta-base
P30	ON-CONTINENT	74.46 %	80.21%
P31	INSTANCE-OF	28.85%	67.35%
P36	HAS-CAPITAL	45.80%	89.76%
P37	HAS-LANGUAGE	30.85%	45.13%
...	...	...	...
P1303	INSTRUMENT	58.69%	100.00%
P1376	CAPITAL-OF	32.05%	81.62%
Mean		31.73%	62.86%

Table 1: Ratio of discarded instances due to multi-token answers in xlm-roberta-base and roberta-base.

Prior work. Various prior works address the issue of predicting multi-subtoken words for single [MASK] tokens (Ghazvininejad et al., 2019; Jiang et al., 2020a; Kalinsky et al., 2023; Shen et al., 2020; Robinson et al., 2023). Jiang et al. (2020a) provided a selection of algorithms to tackle predicting multi-token entities. However, they require a specification of further parameters, such as the number of subtokens to generate. Kalinsky et al. (2023) proposed generation approaches that either require additional training or the use of an external network, making them inapplicable to the purpose of evaluating knowledge contained in pre-trained weights through a zero-shot approach.

2.2 Issue 2: Multiple Correct Answers

LAMA expects exactly one correct answer to each knowledge query and rates other factually correct answers as errors. To illustrate this, consider the query “Germany shares a border with [MASK]”, to which LAMA expects the answer “Poland”. All other correct answers, such as “Denmark” are marked as incorrect. This issue affects all N-M

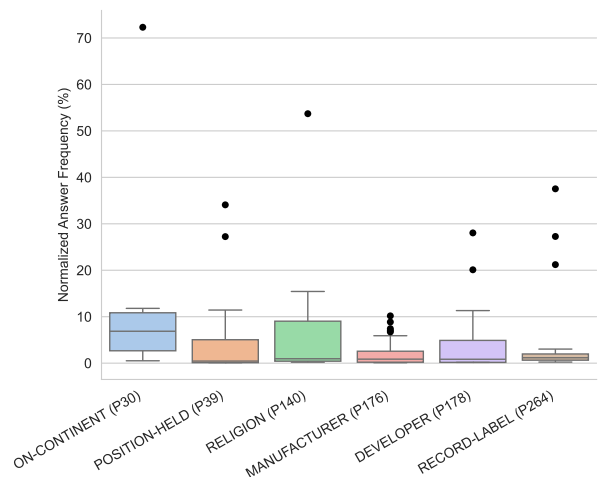


Figure 2: The normalized answer frequency of selected relations in the LAMA probe. The outliers are marked with dots. In some relations, a majority class accounts for more than 50% of all instances.

Template: The capital of [X] is [Y].

Subject: **Uganda**

Answer Options: [**Thimphu**, **Kampala**, **Buenos Aires**, **Bandar Seri Begawan**]

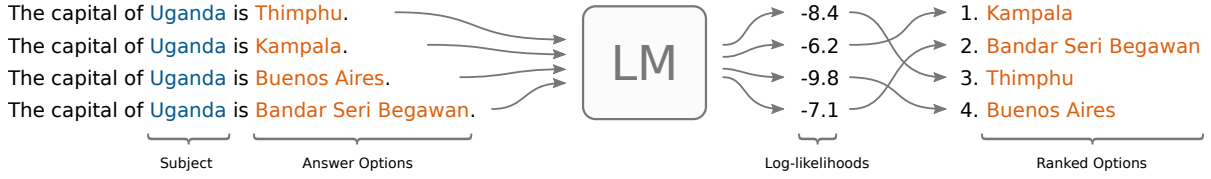


Figure 3: For each answer option, a sentence is passed to the LM (here using the template: “The capital of [X] is [Y].” and the subject “Uganda”). The log-likelihood scores assigned by the LM are then used to rank the answer options.

relations in LAMA.

Prior work. KAMEL (Kalo and Fichtel, 2022) address this by allowing the LM to generate an arbitrary number of answers using a template instructed via few-shot prompting, experimenting with ranges of 1-10 answers per instance. Subsequently, they evaluate the predictions using standard measures of precision and recall. However, their approach relies on the text generation ability of causal LMs and thus cannot be applied to masked LMs.

2.3 Issue 3: Imbalanced Answer Distribution

The relations in T-REx have a highly unbalanced answer distribution (except the 1-1 relations), and in certain relationships, over half of the instances belong to the predominant class (see Figure 2). Zhong et al. (2021) noted that a model that always chooses the majority class outperforms some state-of-the-art LMs on selected relations.

To illustrate, consider the T-REx’s ON-CONTINENT relation, which connects a location to the continent in which it is situated. Counter-intuitively, the majority class in this relation is “Antarctica”, accounting for 72% of all instances.

Prior work. To account for this imbalance, Cao et al. (2021) created a balanced version of the LAMA probe called WikiUNI. It contains the same relations as T-REx but has a uniform answer distribution and was constructed to have the same number of subjects for every object. However, their dataset samples a highly skewed number of instances per relation, with 7 relations (out of 41) accounting for over 50% of all instances.

2.4 Issue 4: Rare Wikidata Entries

The above-mentioned example of “Antarctica” being an answer to over 72% of all instances in the ON-CONTINENT relation also points to another problem: An artifact of randomly sampling Wiki-

data for relation instances is that rare Wikidata entries are overrepresented. For example, ON-CONTINENT has a large number of small islands as subjects (e.g., “Umber Island” and “Brooklyn Island”, both close to the Antarctic continent), many of which are unlikely to occur in a corpus outside of an encyclopedia like Wikipedia. We believe that the LAMA dataset unfairly favors LMs trained using Wikipedia. However, to the best of our knowledge, no prior work has addressed this issue.

2.5 Issue 5: Evaluation of Causal LMs

LAMA relies on the capability of masked language models to fill in masked tokens, making it unsuitable for causal LMs.

Prior work. To address this, Kalo and Fichtel (2022) proposed the KAMEL probe. Factual knowledge is probed by virtue of question statements for which the response is auto-regressively generated using the causal LM. To guide the generation approach, they prepend k few-shot examples into the prompt that present how the correctly formatted answer should look. However, since this approach relies on the language modeling objective of causal LMs, KAMEL does not apply to masked LMs.

Further Related Work

Further related work has attempted to mitigate the sensitivity of the evaluation on the timing of pre-training (and hence the corpus used) (Dhingra et al., 2022; Mallen et al., 2023; Onoe et al., 2022). Nonetheless, as these studies do not directly address the concerns outlined in the previous subsections, we will not discuss them in detail. Interested readers are referred to the cited papers for more information.